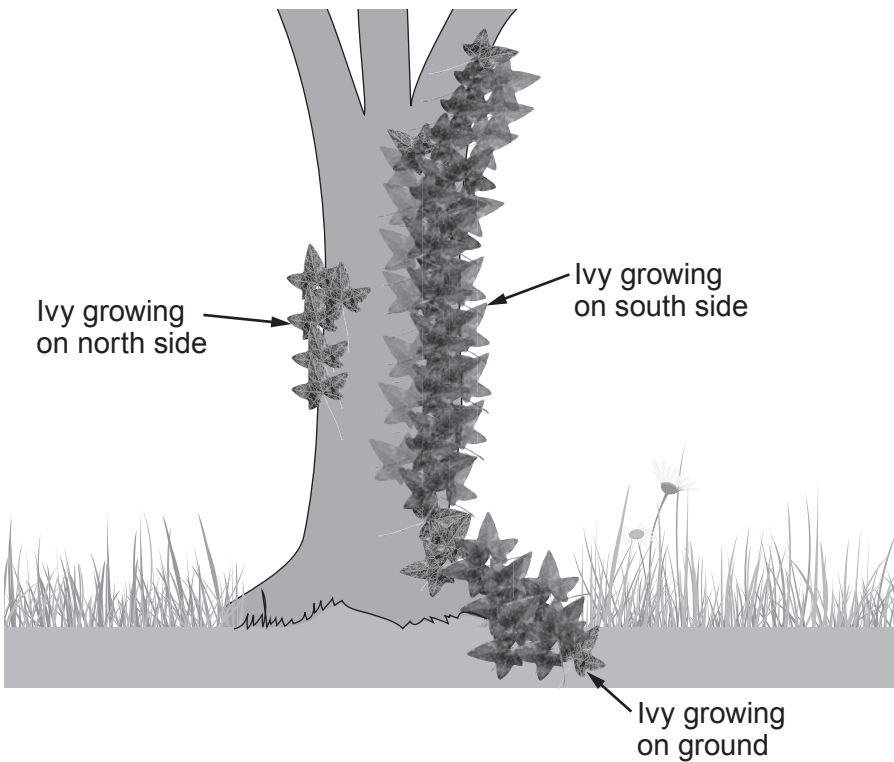
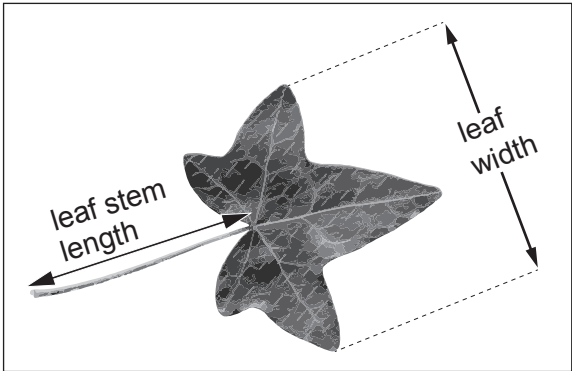


5. Students investigated factors affecting leaf growth of ivy (*Hedera sp.*).

They measured the leaf stem length and the leaf width of 50 leaves of an ivy plant growing on the south side of a tree trunk at a height between one and two metres.



They compared these measurements with ivy leaves of the same ivy plant growing on the north side of the same tree trunk at the same height and also leaves from the same ivy plant growing flat on the ground.



Examiner
only

Their results are shown below:

	Leaves from ivy growing on the south side	Leaves from ivy growing on the north side	Leaves from ivy growing on the ground
Mean leaf stem length (mm)	63	60	42
Mean leaf width (mm)	55	52	38
Ratio of leaf stem length to leaf width	1.15 : 1	 :	1.11 : 1

(a) (i) Calculate the ratio of leaf stem length to leaf width for the ivy growing on the north side of the tree and **write your answer in the table.** [2]
SPACE FOR WORKING

(ii) Suggest why there is a difference between the ratio of leaf stem length to leaf width for the ivy growing on the ground and the ivy growing on the south side of the tree. [1]

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(b) Students recorded the data for the ratio in the frequency table below.

Ratio	South facing	North facing	Ground growing
0.51 - 0.75	7	4	4
0.76 - 1.00	9	7	13
1.01 - 1.25	15	27	24
1.26 - 1.50	13	9	7
1.51 - 1.75	3	5	2
1.76 - 2.00	0	0	0

State the type of variation shown by the ratio. [1]

.....

4

